

Exploratory analysis Flanders double-sampling study

This short report contains my exploratory work with the Flanders double-sampling study and contains some questions on how the data should be used.

Data import and inspection

The dataset contains three csv files.

(1) Sampling points

```
folder_flds <- here("data", "Flanders_Double_Sampling_Data")

# sampling points
sampling_points <- read_delim(
  here(folder_flds,
        "EJPsoil_LUCAS22_Double_sampling_BE_Flanders_Sampling_Points_list.csv"),
  delim = ";",
  col_types = cols(
    # import as characters to fix issues with different notation:
    # both . and , are used as decimal character ???
    WGS84Latitude = col_character(),
    WGS84Longitude = col_character(),
    Elevation = col_character()
  )
)
glimpse(sampling_points)
```

```

Rows: 166
Columns: 11
$ Nb          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
$ LUCAS_PlotID <dbl> 38723090, 38883112, 38923094, 38983120, 39283116, 39323~
$ Town        <chr> "Ronse", "Oosterzele", "Geraardsbergen", "Berlare", "Ka~
$ WGS84Latitude <chr> "50.74562", "50.95486", "50.79672", "51.03398", "51.019~
$ WGS84Longitude <chr> "3.629459", "3.829376", "3.90732", "3.961944", "4.39292~
$ Elevation    <chr> "53,3", "32,6", "15,3", "2,8", "6,7", "4,6", "4,4", "3,~
$ FieldObs_LU  <chr> "Woodland", "Woodland", "Woodland", "Woodland", "Woodla~
$ LUCAS_LU     <chr> "Woodland", "Woodland", "Woodland", "Woodland", "Woodla~
$ RSG          <chr> "Retisols", "Stagnosols", "Cambisols", "Phaeozems", "Ph~
$ Bio_Point    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ LUCAS2018    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0~

```

Little data cleaning and formatting as `sf` object:

```

sampling_points_clean <- sampling_points %>%
  mutate(
    across(c(WGS84Latitude, WGS84Longitude, Elevation),
      ~ as.numeric(gsub(",", ".", .x)))
  ) %>%
  st_as_sf(coords = c("WGS84Longitude", "WGS84Latitude"),
    crs = "WGS84") %>%
  select(LUCAS_PlotID, Town, Elevation, contains("LU"), RSG, Bio_Point)

glimpse(sampling_points_clean)

```

```

Rows: 166
Columns: 9
$ LUCAS_PlotID <dbl> 38723090, 38883112, 38923094, 38983120, 39283116, 3932311~
$ Town        <chr> "Ronse", "Oosterzele", "Geraardsbergen", "Berlare", "Kape~
$ Elevation    <dbl> 53.3, 32.6, 15.3, 2.8, 6.7, 4.6, 4.4, 3.9, 31.0, 13.9, 32~
$ FieldObs_LU  <chr> "Woodland", "Woodland", "Woodland", "Woodland", "Woodland~
$ LUCAS_LU     <chr> "Woodland", "Woodland", "Woodland", "Woodland", "Woodland~
$ LUCAS2018    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ~
$ RSG          <chr> "Retisols", "Stagnosols", "Cambisols", "Phaeozems", "Phae~
$ Bio_Point    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ geometry     <POINT [°]> POINT (3.629459 50.74562), POINT (3.829376 50.95486~

```

Five data points lie in Wallonia and 1 in Brussels. Question: Do we need to retain them for the comparison of sampling schemes (LUCAS vs. Cmon)?

```

# import provinces of Flanders
ogc_vrbg <- "https://geo.api.vlaanderen.be/VRBG2025/ogc/features/v1/"
provinces <- get_feature_ogc(
  url = ogc_vrbg,
  collection = "Refprv",
  crs = "EPSG:31370"
) %>% st_transform(crs = "WGS84")

# print points outside Flanders
outside_idx <- st_disjoint(
  sampling_points_clean,
  st_union(provinces),
  sparse = FALSE
)
sampling_points_clean[outside_idx, ] %>%
  st_drop_geometry() %>%
  select(Town) %>%
  print()

```

```

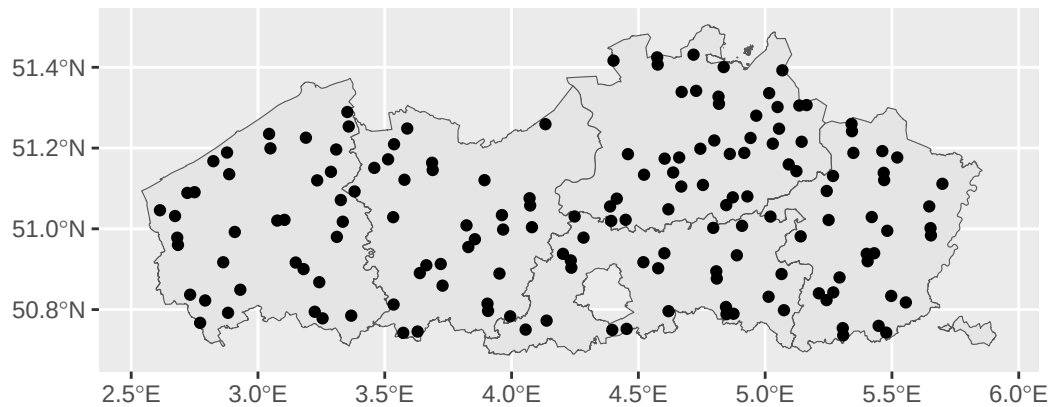
# A tibble: 6 x 1
  Town
  <chr>
1 Sint-Jans-Molenbeek
2 Cerfontaine
3 Sombreffe
4 Seneffe
5 Genappe
6 Burdinne

```

```

# filter on points inside Flanders (N = 160)
sampling_points_fl <- st_filter(
  sampling_points_clean,
  provinces
)
ggplot() +
  geom_sf(data = provinces) +
  geom_sf(data = sampling_points_fl)

```



(2) Soildata

```
soildata <- read_csv2(
  here(folder_flds,
    "EJPsoil_LUCAS22_Double_sampling_BE_Flanders_Soildata.csv")
)
glimpse(soildata)
```

Rows: 559

Columns: 22

```
$ LUCAS_PlotID <dbl> 38043130, 38043130, 38043130, 38083120, 38083120, 380831~
$ FieldSampleID <chr> "38043130_0-10", "38043130_10-30", "38043130_LUCAS", "38~
$ LabSampleID <chr> "23-003424", "23-003425", "23-003426", "23-003430", "23-~
~
$ SampScheme <chr> "Cmon", "Cmon", "LUCAS", "Cmon", "Cmon", "LUCAS", "Cmon"~
$ Matrix <chr> "Mineral", "Mineral", "Mineral", "Mineral", "Mineral", "Mineral", "~
$ Depth <chr> "0-10", "okt/30", "0-30", "0-10", "okt/30", "0-30", "0-~
1~
$ Thickness <dbl> 10, 20, 30, 10, 20, 30, 10, 20, 30, 10, 20, 30, 10, 20, ~
$ Sand <dbl> 38.20, 40.30, 43.30, 39.90, 22.80, 40.90, 34.40, 37.20, ~
$ Silt <dbl> 41.00, 42.70, 43.10, 46.70, 67.90, 46.70, 50.10, 49.70, ~
```

```

$ Clay      <dbl> 20.80, 17.00, 13.60, 13.30, 9.30, 12.40, 15.60, 13.20, 1~
$ TC        <dbl> 18.7, 19.3, 19.6, 13.7, 13.2, 14.7, 29.9, 12.8, 21.2, 28~
$ TIC       <dbl> 7.21, 7.24, 7.25, NA, NA, NA, NA, NA, 9.19, 8.27, 14~
$ TOC       <dbl> 11.46, 12.03, 12.36, 13.68, 13.16, 14.70, 29.94, 12.80, ~
$ TN        <dbl> 1.21, 1.25, 1.30, 1.49, 1.45, 1.61, 3.04, 1.34, 2.16, 1.~
$ pH_KCl_v_v <dbl> 7.61, 7.54, 7.69, 5.84, 5.13, 6.14, 4.80, 4.60, 4.29, 7.~
$ EC_m_v    <dbl> 116.67, 126.09, 111.66, NA, NA, NA, NA, NA, NA, 149.51, ~
$ EC_v_v    <dbl> 96.95, 104.83, 98.67, NA, NA, NA, NA, NA, NA, 136.87, 15~
$ pH_H2O_m_v <dbl> 8.39, 8.36, 8.38, NA, NA, NA, NA, NA, NA, 8.34, 8.38, 8.~
$ pH_H2O_v_v <dbl> 8.32, 8.27, 8.26, NA, NA, NA, NA, NA, NA, 8.20, 8.38, 8.~
$ pH_KCl_m_v <dbl> 7.51, 7.48, 7.59, NA, NA, NA, NA, NA, NA, 7.54, 7.59, 7.~
$ pH_CaCl2_m_v <dbl> 7.600, 7.570, 7.410, NA, NA, NA, NA, NA, NA, 7.650, 7.69~
$ pH_CaCl2_v_v <dbl> 7.49, 7.46, 7.48, NA, NA, NA, NA, NA, NA, 7.50, 7.68, 7.~

```

Little data cleaning step and filtering observations. Questions/remarks: Do we need to retain the data from the strooisel-laag? There is 1 plot with 6 observations, are these partial duplicates? There are two points that are listed in the previous csv file but have no corresponding records in the soildata csv.

```

# Depth data were formatted as a date (probably a setting from Excel/GSheet?)
soildata_clean <- soildata %>%
  mutate(
    Depth = case_when(Depth == "okt/30" ~ "10-30",
                      TRUE ~ Depth)
  )

```

```

# 136 plots have 3 observations (2*Cmon + 1*LUCAS), 29 plots have 5 observations (mineral + c
soildata_clean %>%
  summarise(.by = LUCAS_PlotID, N = n()) %>%
  summarise(.by = N, plots = n())

```

```

# A tibble: 3 x 2
      N plots
  <int> <int>
1     3   136
2     5    29
3     6     1

```

```

# selecting soildata: remove strooisel-laag, keep only points in Flanders and remove partial
soildata_clean_fl <- soildata_clean %>%
  filter(

```

```

!grepl("strooisel", tolower(FieldSampleID)),
  LUCAS_PlotID %in% sampling_points_fl$LUCAS_PlotID
) %>%
distinct(LUCAS_PlotID, Depth, .keep_all = T)

# there are 2 points that appear in the sampling_points object but that don't appear in the s
sampling_points_fl %>%
  filter(!(LUCAS_PlotID %in% soildata_clean_fl$LUCAS_PlotID)) %>%
  st_drop_geometry() %>%
  select(LUCAS_PlotID, Town) %>%
  print()

```

```

# A tibble: 2 x 2
  LUCAS_PlotID Town
      <dbl> <chr>
1    38463096 Kortrijk
2    39743098 Kortenaken

```

Question: Why are some TOC values sometimes larger than TC? (is TC not the sum of TOC and TIC?; is there a rounding issue?) Missing values in TIC when TC and TOC are both observed, can you just use TC - TOC as imputed value?

```

soildata_clean_fl %>%
  mutate(
    TC = janitor::round_half_up(TC, 1),
    TOC = janitor::round_half_up(TOC, 1),
    TIC = janitor::round_half_up(TIC, 1)
  ) %>%
  filter(TOC > TC) %>%
  select(LUCAS_PlotID, TC, TOC, TIC)

```

```

# A tibble: 16 x 4
  LUCAS_PlotID    TC    TOC    TIC
      <dbl> <dbl> <dbl> <dbl>
1    38363144  25.8  25.9    NA
2    38363148  21.6  21.7    NA
3    38703142  24.9  25     NA
4    39323116  31.9  32     NA
5    39423090    7    7.1    NA
6    39463128  13.5  13.6    NA
7    39503150  18.3  18.4    NA

```

8	39823134	31.2	31.3	NA
9	39823144	17	17.1	NA
10	39963138	20.1	20.2	NA
11	40043090	28	28.1	NA
12	40043090	15.9	16	NA
13	40043108	12.1	12.2	NA
14	40043130	56	56.1	NA
15	40163108	19.1	19.2	NA
16	40203120	53.3	53.4	NA

Question: To compare LUCAS vs. Cmon, should we combine the two Cmon measurements? Can you take the weighted average of the 0-10 cm sample and the 10-30 cm sample (with respective weights 1/3 and 2/3) to get an average for the 0-30 cm depth?

```
# weighted average for Cmon sampling scheme:
soildata_clean_fl_avg <- soildata_clean_fl %>%
  mutate(
    weights = case_when(Depth == "0-10" ~ 1/3,
                        Depth == "10-30" ~ 2/3,
                        Depth == "0-30" ~ 1),
    .after = Depth
  ) %>%
  summarise(
    .by = c(LUCAS_PlotID, SampScheme, Matrix),
    Depth = "0-30",
    Thickness = sum(Thickness),
    across(c(Sand, Silt, Clay, TC, TIC, TOC, TN,
              contains("pH"), contains("EC")),
            ~ weighted.mean(.x, w = weights, na.rm = T))
  ) %>%
  mutate(
    across(c(Sand, Silt, Clay, TC, TIC, TOC, TN,
              contains("pH"), contains("EC")),
            ~ replace_when(.x, is.nan(.x) ~ NA_real_))
  )
```

(3) Bulk density

```
bulkdens <- read_csv2(
  here(folder_flds,
```

```

      "EJPsoil_LUCAS22_Double_sampling_BE_Flanders_Bulk_density.csv")
    )
  glimpse(bulkdens)

```

```

Rows: 331
Columns: 8
$ LUCAS_PlotID  <dbl> 29723114, 29723114, 38043130, 38043130, 38083120, 380831~
$ LayerID      <chr> "29723114_0-10 cm", "29723114_10-30 cm", "38043130_0-
10 ~
$ Depth        <chr> "0-10", "10-30", "0-10", "10-30", "0-10", "10-30", "0-
10~
$ BD_mean      <chr> "1.55", "1.63", "1.51", "1.50", "1.44", "1.50", "1.35", ~
$ BD_sd        <chr> "0.087", "0.075", "0.014", "0.053", "0.035", "0.013", "0~
$ BD_CV        <chr> "5.65", "4.62", "0.94", "3.50", "2.43", "0.84", "6.45", ~
$ SWC_volp_mean <dbl> 1373, 1325, 4698, 4778, 4765, 4605, 4243, 3978, 5175, 51~
$ SWC_volp_sd  <chr> "2.31", "1.59", "1.29", "2.10", "0.50", "0.89", "3.59", ~

```

Little data cleaning step and weighted-average:

```

bulkdens_clean <- bulkdens %>%
  mutate(
    across(c(contains("BD_"), contains("SWC_")), ~ as.numeric(.x))
  ) %>%
  mutate(
    weights = case_when(Depth == "0-10" ~ 1/3,
                        Depth == "10-30" ~ 2/3,
                        Depth == "0-30" ~ 1),
    .after = Depth
  ) %>%
  summarise(
    .by = LUCAS_PlotID,
    Depth = "0-30",
    SampScheme = "Cmon",
    across(c(contains("BD_"), contains("SWC_")),
      ~ weighted.mean(.x, w = weights, na.rm = T))
  )

```

Combine datasets and some further cleaning


```

soildata_flds <- soildata_clean_fl_avg %>%
  left_join(sampling_points_fl, by = "LUCAS_PlotID") %>%
  left_join(bulkdens_clean, by = c("LUCAS_PlotID", "Depth", "SampScheme")) %>%
  select(
    LUCAS_PlotID, Town, Elevation, geometry, LUCAS_LU, FieldObs_LU, RSG,
    #FieldSampleID, LabSampleID,
    SampScheme, Depth, Matrix,
    Thickness, Sand, Silt, Clay, TC, TIC, TOC, TN,
    contains("pH_"), contains("EC_"),
    #LayerID,
    BD_mean, BD_sd, BD_CV, SWC_volp_mean, SWC_volp_sd
  ) %>%
  mutate(
    Thickness = set_units(Thickness, "cm"),
    Elevation = set_units(Elevation, "m"),
    Sand = set_units(Sand, "%"),
    Silt = set_units(Silt, "%"),
    Clay = set_units(Clay, "%"),
    TC = set_units(TC, "g/kg"),
    TIC = set_units(TIC, "g/kg"),
    TOC = set_units(TOC, "g/kg"),
    TN = set_units(TN, "g/kg"),
    EC_m_v = set_units(EC_m_v, "µS/cm"),
    EC_v_v = set_units(EC_v_v, "µS/cm"),
    BD_mean = set_units(BD_mean, "g/cm³"),
    BD_sd = set_units(BD_sd, "g/cm³"),
    BD_CV = set_units(BD_CV, "g/cm³"),
    SWC_volp_mean = set_units(SWC_volp_mean, "%"),
    SWC_volp_sd = set_units(SWC_volp_sd, "%")
  )

summary(soildata_flds)

```

LUCAS_PlotID	Town	Elevation	geometry
Min. :38043130	Length :316	Min. : 0.90	POINT :316
1st Qu.:38683138	N.unique :123	1st Qu.: 10.60	epsg:4326 : 0
Median :39373116	N.blank : 0	Median : 24.35	+proj=long...: 0
Mean :39229448	Min.nchar: 3	Mean : 31.07	
3rd Qu.:39743148	Max.nchar: 21	3rd Qu.: 44.20	
Max. :40223060		Max. :123.00	

LUCAS_LU	FieldObs_LU	RSG	SampScheme
Length :316	Length :316	Length :316	Length :316
N.unique : 5	N.unique : 5	N.unique : 13	N.unique : 2
N.blank : 0	N.blank : 0	N.blank : 0	N.blank : 0
Min.nchar: 7	Min.nchar: 7	Min.nchar: 7	Min.nchar: 4
Max.nchar: 10	Max.nchar: 9	Max.nchar: 10	Max.nchar: 5

Depth	Matrix	Thickness	Sand	Silt
Length :316	Length :316	Min. :30	Min. : 3.60	Min. : 1.70
N.unique : 1	N.unique : 1	1st Qu.:30	1st Qu.:19.65	1st Qu.:15.53
N.blank : 0	N.blank : 0	Median :30	Median :49.78	Median :33.38
Min.nchar: 4	Min.nchar: 7	Mean :30	Mean :49.53	Mean :36.32
Max.nchar: 4	Max.nchar: 7	3rd Qu.:30	3rd Qu.:77.55	3rd Qu.:56.82
		Max. :30	Max. :95.50	Max. :78.90

Clay	TC	TIC	TOC
Min. : 1.967	Min. : 2.40	Min. : 0.1000	Min. : 2.37
1st Qu.: 6.638	1st Qu.: 13.62	1st Qu.: 0.1567	1st Qu.:13.46
Median :13.483	Median : 19.10	Median : 0.5500	Median :18.94
Mean :14.141	Mean : 23.30	Mean : 3.1703	Mean :22.56
3rd Qu.:18.775	3rd Qu.: 26.65	3rd Qu.: 2.5600	3rd Qu.:26.24
Max. :49.200	Max. :134.50	Max. :48.8300	Max. :88.68

TN	pH_KCl_v_v	pH_H2O_m_v	pH_H2O_v_v
Min. :0.140	Min. :2.750	Min. :3.830	Min. :3.800
1st Qu.:1.133	1st Qu.:4.310	1st Qu.:4.860	1st Qu.:4.910
Median :1.467	Median :5.083	Median :6.055	Median :6.080
Mean :1.801	Mean :5.208	Mean :6.031	Mean :5.999
3rd Qu.:2.133	3rd Qu.:6.140	3rd Qu.:7.114	3rd Qu.:7.120
Max. :8.370	Max. :7.940	Max. :8.440	Max. :8.360
		NAs :241	
		NAs :174	NAs :173

pH_KCl_m_v	pH_CaCl2_m_v	pH_CaCl2_v_v	EC_m_v
Min. :2.690	Min. :3.185	Min. :2.900	Min. : 23.98
1st Qu.:3.840	1st Qu.:4.044	1st Qu.:4.065	1st Qu.: 52.95
Median :5.030	Median :5.280	Median :5.330	Median : 83.56
Mean :5.088	Mean :5.296	Mean :5.294	Mean : 110.85
3rd Qu.:6.282	3rd Qu.:6.383	3rd Qu.:6.450	3rd Qu.: 117.51
Max. :7.770	Max. :7.820	Max. :7.790	Max. :1026.34
NAs :180	NAs :174	NAs :173	NAs :173

EC_v_v	BD_mean	BD_sd	BD_CV
Min. : 19.01	Min. :0.490	Min. :0.01200	Min. : 0.7933
1st Qu.: 42.39	1st Qu.:1.283	1st Qu.:0.05967	1st Qu.: 4.2300

Median :	63.62	Median :	1.407	Median :	0.08667	Median :	6.4467
Mean :	93.91	Mean :	1.359	Mean :	0.10018	Mean :	8.2955
3rd Qu.:	95.72	3rd Qu.:	1.483	3rd Qu.:	0.12333	3rd Qu.:	9.7933
Max. :	1093.39	Max. :	1.703	Max. :	0.44667	Max. :	77.5433
NAs :	174	NAs :	163	NAs :	163	NAs :	163
SWC_volp_mean		SWC_volp_sd					
Min. :	-2495	Min. :	0.500				
1st Qu.:	1681	1st Qu.:	1.647				
Median :	2773	Median :	2.710				
Mean :	2819	Mean :	4.125				
3rd Qu.:	3793	3rd Qu.:	4.477				
Max. :	7194	Max. :	83.997				
NAs :	163	NAs :	163				

Data exploration

Landuse & soil types

Subgroup analyses within land-use and soil types will need to be restricted to groups with enough observations.

```
soildata_flds %>%
  summarise(.by = FieldObs_LU, N = n_distinct(LUCAS_PlotID)) %>%
  arrange(desc(N)) %>%
  print(n = 5)
```

```
# A tibble: 5 x 2
  FieldObs_LU      N
  <chr>         <int>
1 Cropland         61
2 Grassland        51
3 Woodland         41
4 Wetland           4
5 Orchard           1
```

```
soildata_flds %>%
  summarise(.by = RSG, N = n_distinct(LUCAS_PlotID)) %>%
  arrange(desc(N)) %>%
  print(n = 15)
```

```
# A tibble: 13 x 2
  RSG      N
  <chr>   <int>
1 Cambisols    41
2 Luvisols    23
3 Retisols    19
4 Anthrosols   19
5 Podzols     17
6 Umbrisols     8
7 Phaeozems     7
8 Stagnosols     6
9 Arenosols     6
10 Gleysols     5
11 Technosols    3
12 Planosols     2
13 Regosols     2
```

Comparing LUCAS & Cmon sampling schemes

The graphs below show a scatter plot and loess smoother for the parameter measured with LUCAS vs. Cmon (weighted average). The bisector is added for reference. Detection of bivariate outliers is performed with the robust method Minimum Covariance Determinant (performance package). Not all the detected outliers will be influential for model fitting but may signal interesting differences between LUCAS and Cmon. Alternative: outlier detection with Bland-Altman analysis.

```
# plot + outlier detection with Minimum Covariance Determinant
compare_schemes <- function(vrb) {
  data_wide <- soildata_flds %>%
    select(LUCAS_PlotID, SampScheme, all_of(vrb)) %>%
    pivot_wider(names_from = SampScheme, values_from = all_of(vrb),
                names_prefix = paste0(vrb, "_")) %>%
    filter(
      !is.na(.data[[paste0(vrb, "_Cmon")]]),
      !is.na(.data[[paste0(vrb, "_LUCAS")]])
    )
  outl <- performance::check_outliers(
    data_wide %>%
      select(-LUCAS_PlotID),
    method = "mcd", percentage_central = .75,
    verbose = TRUE
  )
}
```

```

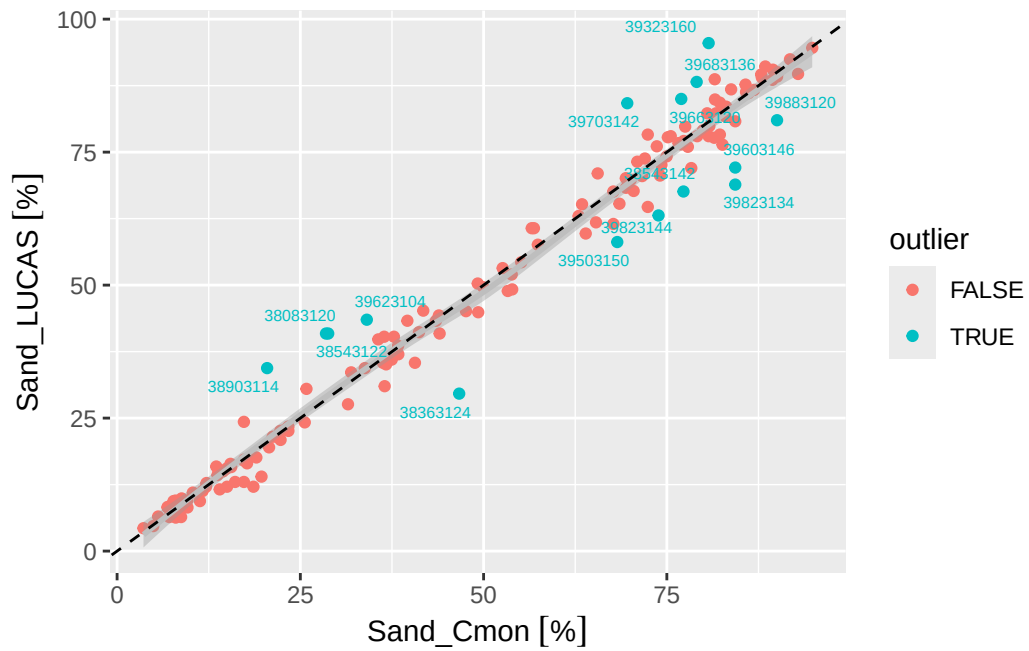
data_wide$outlier <- FALSE
data_wide$outlier[which(outl)] <- TRUE

data_wide %>%
  mutate(
    lab_outl = case_when(outlier ~ LUCAS_PlotID, TRUE ~ NA_real_)
  ) %>%
  ggplot(aes(.data[[paste0(vrb,"_Cmon")]],
             .data[[paste0(vrb,"_LUCAS")]],
             color = outlier)) +
  geom_point() +
  geom_smooth(
    inherit.aes = F,
    aes(.data[[paste0(vrb,"_Cmon")]],
        .data[[paste0(vrb,"_LUCAS")]]),
    method = "loess", color = "grey"
  ) +
  ggrepel::geom_text_repel(aes(label = lab_outl), show.legend = F,
                           size = 2) +
  geom_abline(slope = 1, intercept = 0, lty = 2)
}

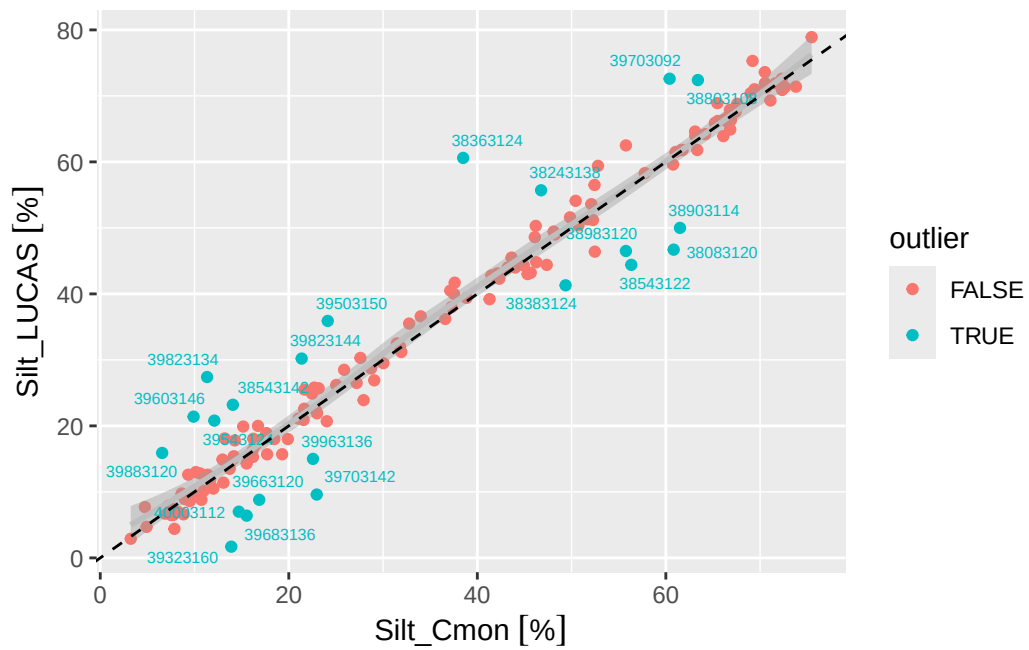
# apply function
sapply(
  FUN = compare_schemes,
  X = c("Sand","Silt","Clay","TC","TOC","TIC","TN",
        "pH_KCl_v_v","pH_KCl_m_v",
        "pH_H2O_m_v","pH_H2O_v_v",
        "pH_CaCl2_m_v","pH_CaCl2_v_v",
        "EC_m_v","EC_v_v")
)

```

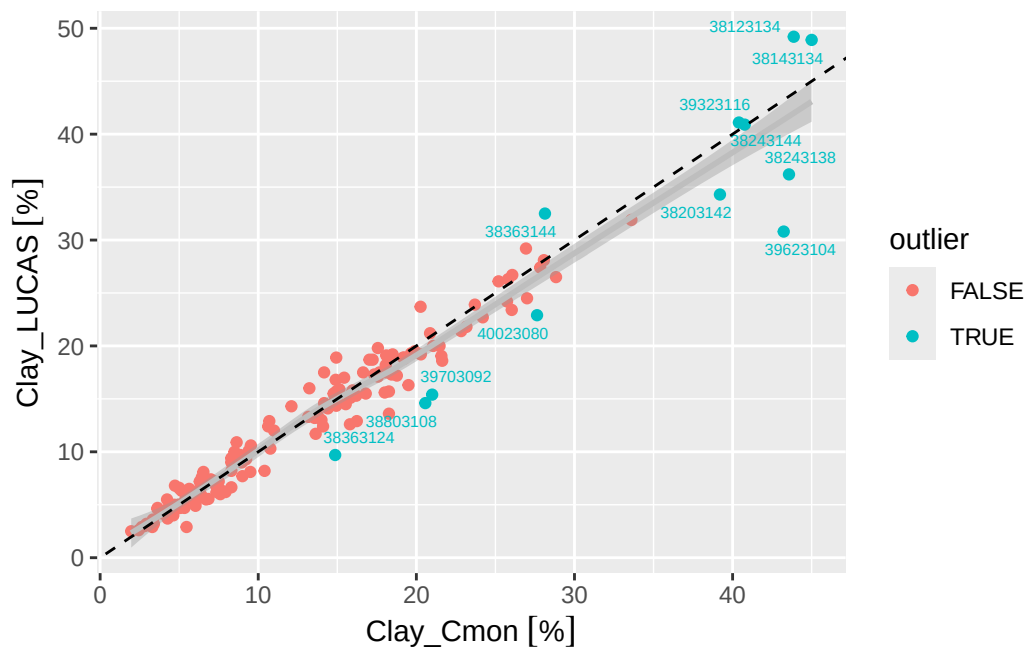
\$Sand



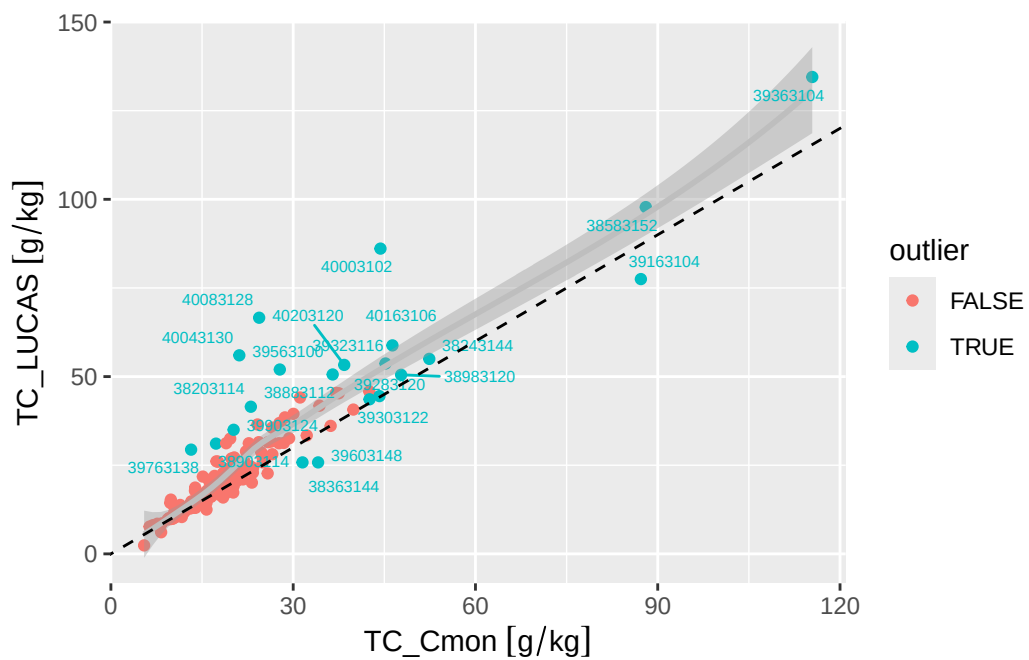
Silt



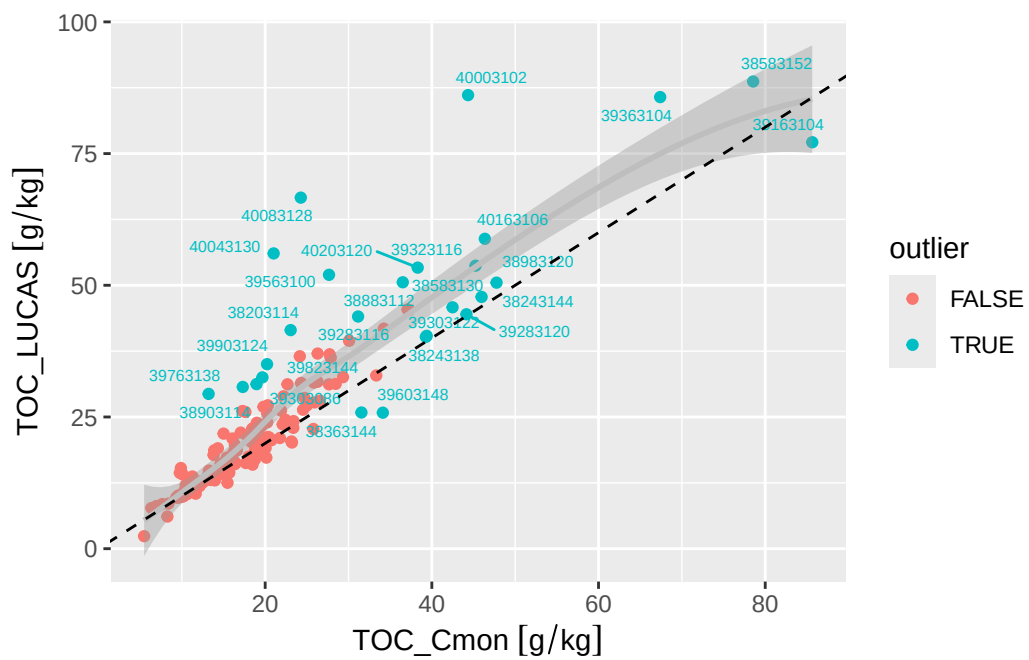
\$Clay



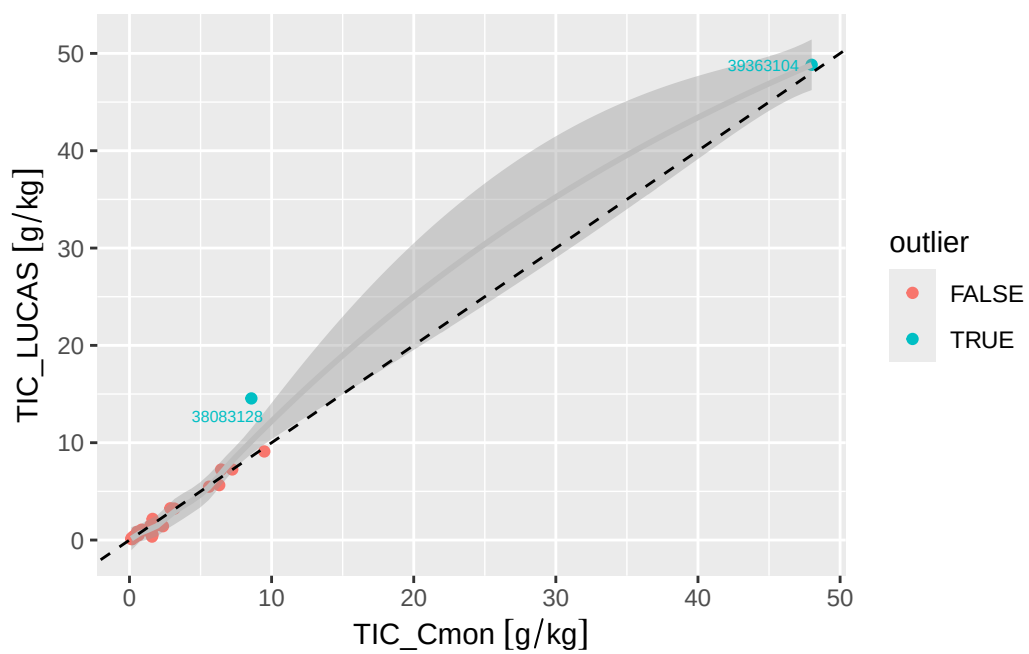
\$TC



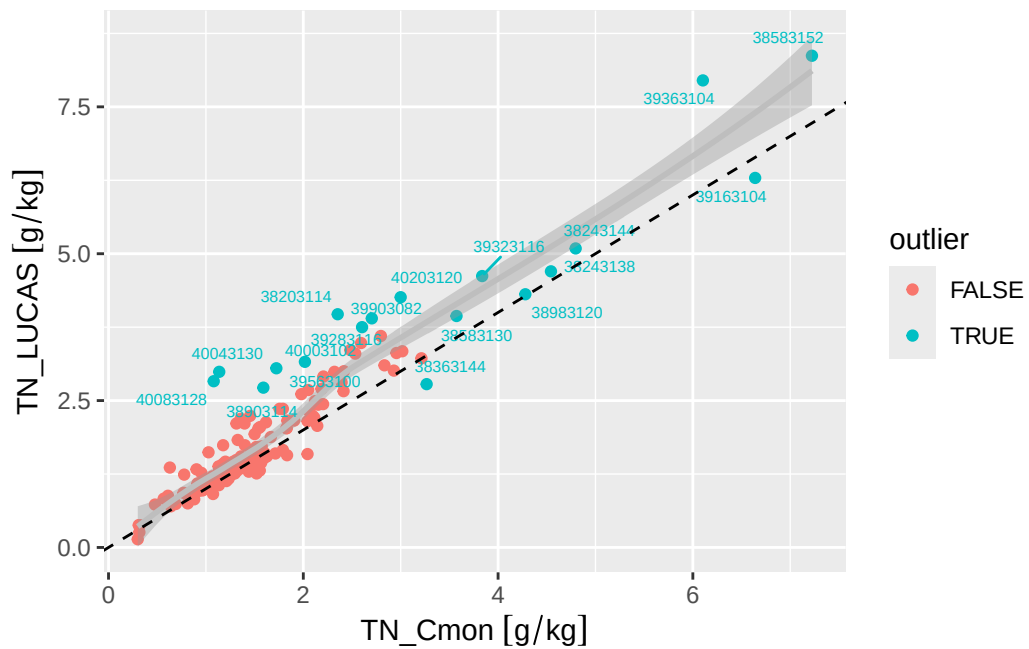
\$TOC



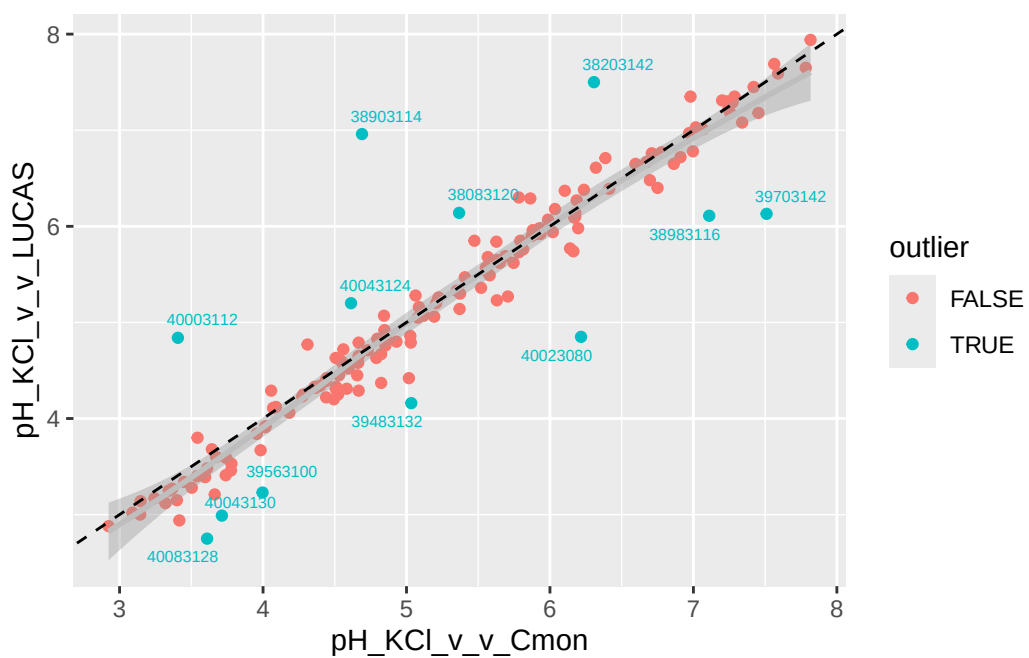
\$TIC



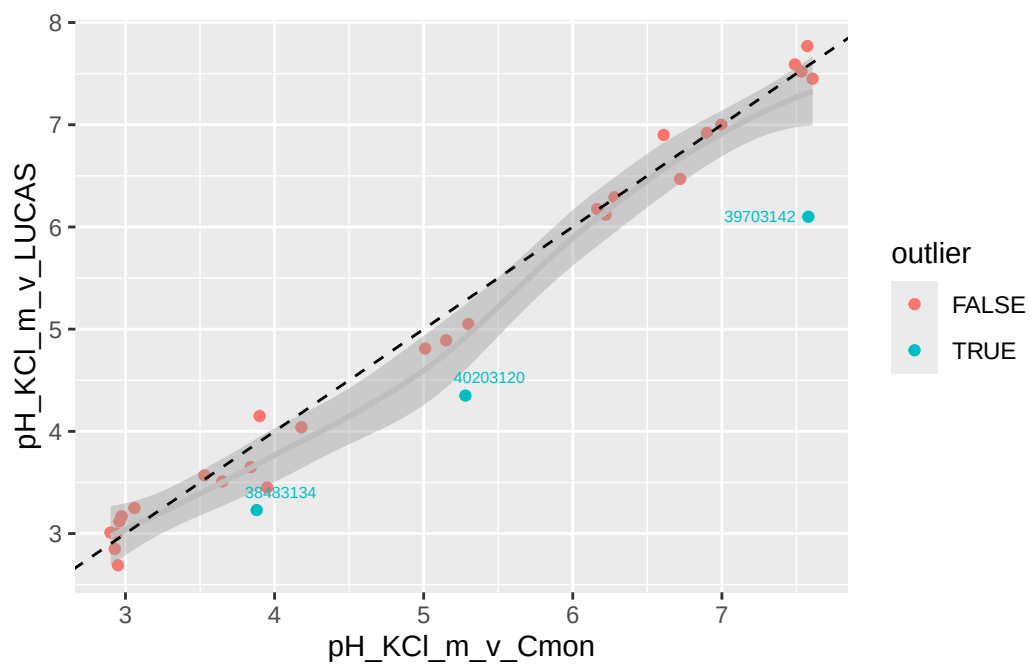
\$TN



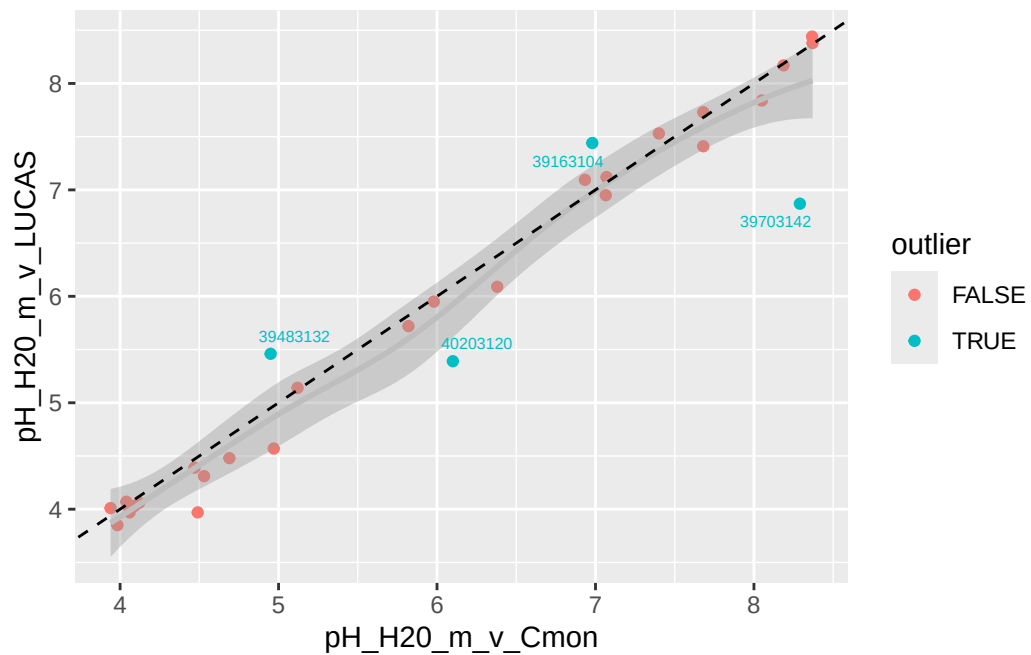
$pH_KCl_v_v$



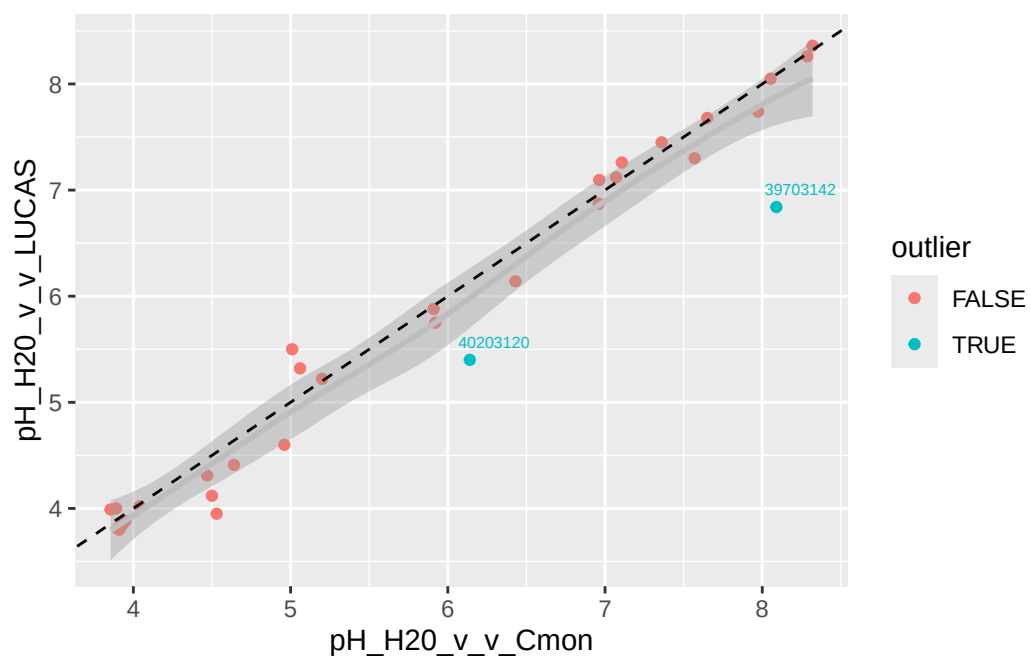
\$pH_KCl_m_v



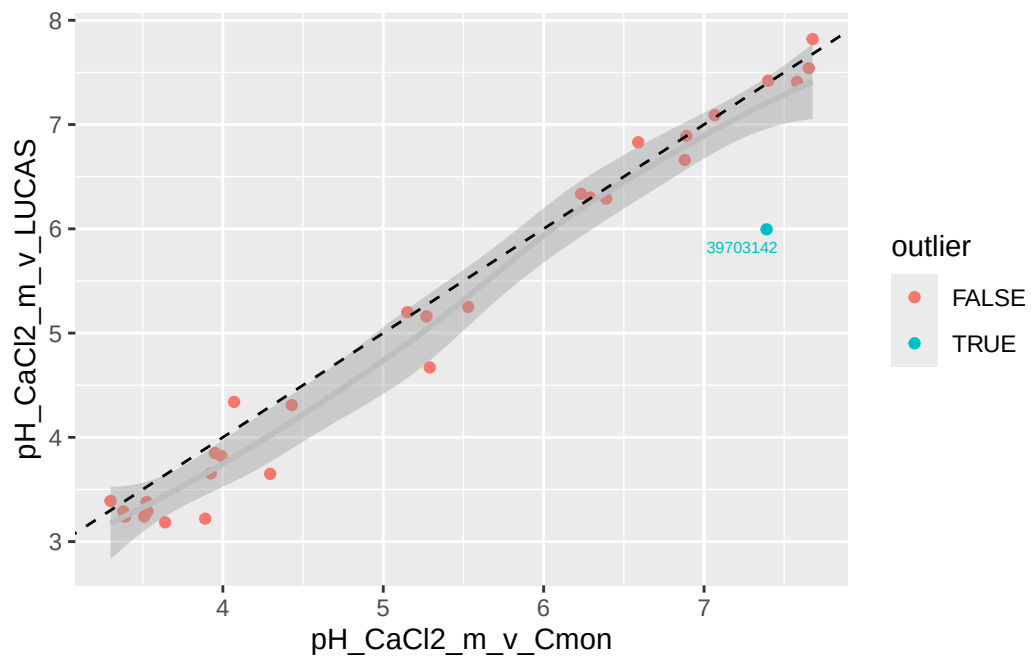
\$pH_H2O_m_v



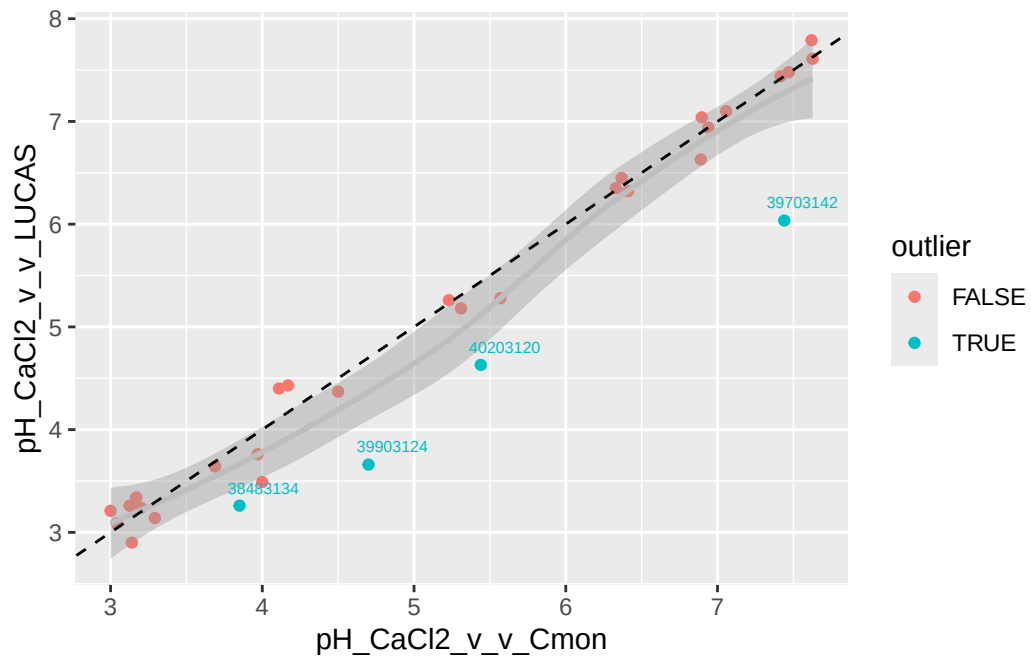
\$pH_H2O_v_v



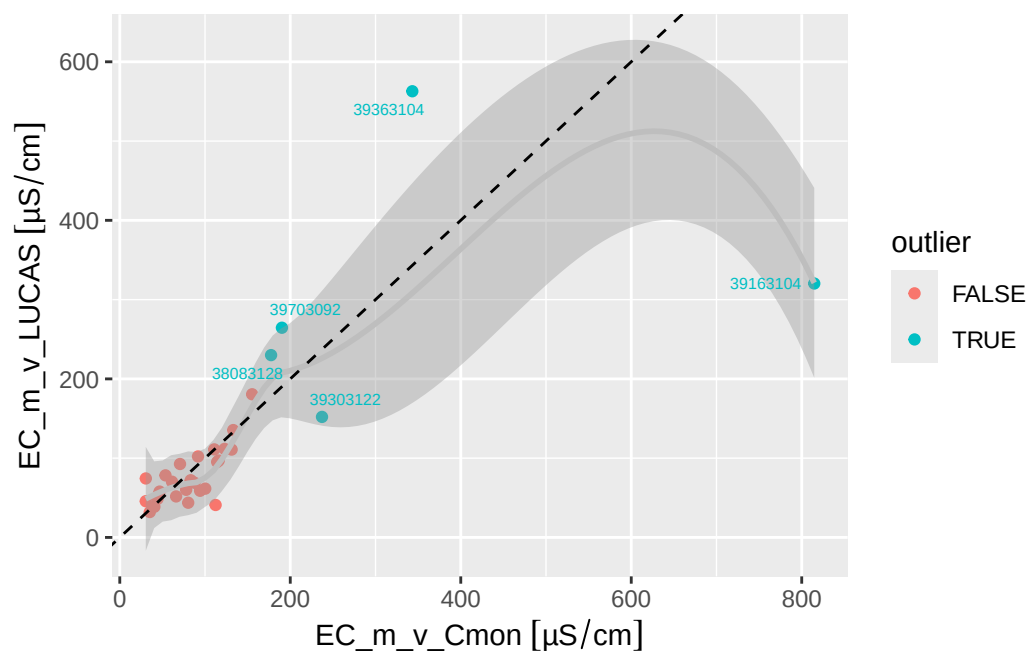
\$pH_CaCl2_m_v



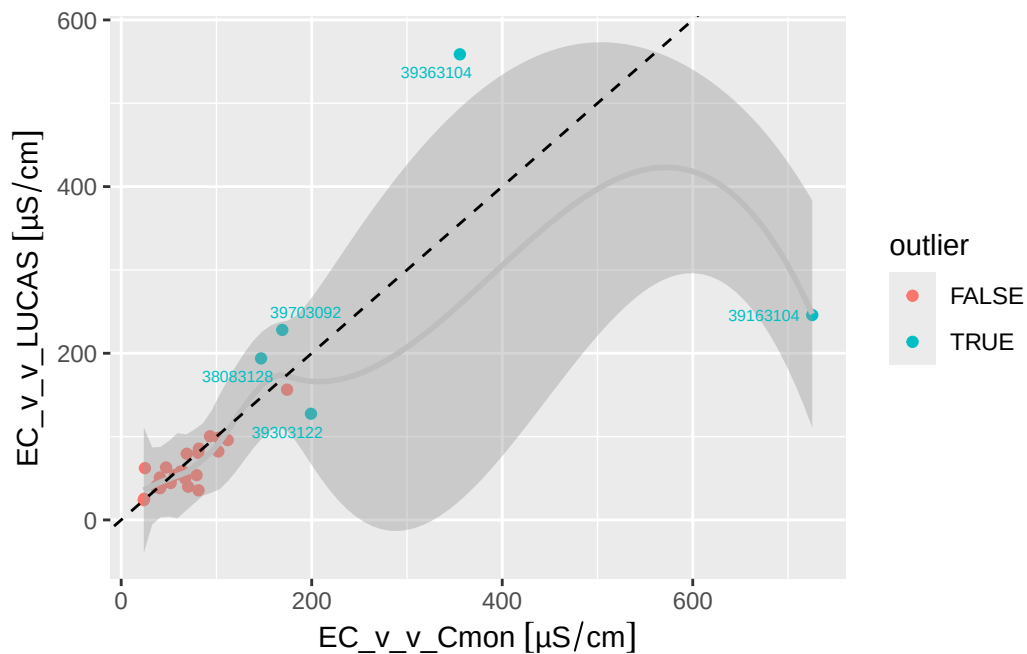
\$pH_CaCl2_v_v



EC_m_v



\$EC_v_v



Question: For the comparison between Cmon and LUCAS; should a TF be developed to translate (predict) values between them or are descriptive statistics enough? Eg. Bland-Altman, Wilcox-test.

```
# to be decided: statistical comparison with a Wilcox test (location-shift)
# => need to apply a multiple-comparison correction (15 variables tested)
compare_schemes_wilcox <- function(vrb) {
  data_wide <- soildata_flds %>%
    select(LUCAS_PlotID, SampScheme, all_of(vrb)) %>%
    mutate(across(all_of(vrb), ~ as.numeric(.x))) %>%
    pivot_wider(names_from = SampScheme, values_from = all_of(vrb),
                names_prefix = paste0(vrb, "_")) %>%
    filter(
      !is.na(.data[[paste0(vrb, "_Cmon")]]),
      !is.na(.data[[paste0(vrb, "_LUCAS")]])
    )
  wilcox.test(
    data_wide[[paste0(vrb, "_Cmon")]],
    data_wide[[paste0(vrb, "_LUCAS")]],
    paired = T, conf.int = T, correct = T,

```

```

) %>% broom::tidy()
}

# apply function
sapply(
  FUN = compare_schemes_wilcox,
  X = c("Sand", "Silt", "Clay", "TC", "TOC", "TIC", "TN",
        "pH_KCl_v_v", "pH_KCl_m_v",
        "pH_H2O_m_v", "pH_H2O_v_v",
        "pH_CaCl2_m_v", "pH_CaCl2_v_v",
        "EC_m_v", "EC_v_v")
) %>% t() %>%
  as.data.frame() %>%
  select(estimate, p.value, conf.low, conf.high) %>%
  # FDR correction
  mutate(q.value = p.adjust(p.value, method = "BH")) %>%
  print(digits = 2)

```

	estimate	p.value	conf.low	conf.high	q.value
Sand	0.2	0.33	-0.22	0.67	4.2e-01
Silt	-0.42	0.045	-0.87	-5.6e-05	9.7e-02
Clay	0.15	0.24	-0.1	0.38	3.6e-01
TC	-2.2	2.4e-13	-3.1	-1.5	1.8e-12
TOC	-2.2	5.6e-13	-3.1	-1.5	2.8e-12
TIC	0.02	0.82	-0.18	0.21	8.2e-01
TN	-0.18	1.8e-14	-0.25	-0.13	2.6e-13
pH_KCl_v_v	0.057	0.00029	0.028	0.09	1.1e-03
pH_KCl_m_v	0.07	0.16	-0.016	0.19	2.6e-01
pH_H2O_m_v	0.075	0.045	0.0025	0.16	9.7e-02
pH_H2O_v_v	0.08	0.059	-0.0017	0.18	1.1e-01
pH_CaCl2_m_v	0.13	0.0025	0.048	0.23	7.4e-03
pH_CaCl2_v_v	0.055	0.27	-0.027	0.21	3.7e-01
EC_m_v	4.4	0.5	-8.6	17	5.4e-01
EC_v_v	2.3	0.5	-5.3	11	5.4e-01

Transfer functions pH

For this analysis we don't need the averaged Cmon data because we need to compare methods that are measured at each depth, but depth could play the role of a covariate if necessary. Strong correlations ($\geq .98$) are observed between all pairs of pH measurement methods (same when Spearman instead of Pearson).


```
soildata_clean_fl %>%
  select(contains("pH")) %>%
  GGally::ggscatmat(corMethod = c("pearson"))
```

